

Experiment - 8

Generation of Mathematical Series Using a Menu-Driven C Program

Write a menu-driven C program to perform one of the following operations using a switch statement:

1. Print an arithmetic progression: Input the first term a , common difference d , and number of terms n .
 $a, a + d, a + 2d, \dots$
2. Print a geometric progression: Input the first term a , common ratio r , and number of terms n .
 $a, ar, ar^2, \dots$
3. Print alternating positive and negative integers: Input the number of terms n .
 $1 -2 3 -4 5 -6 \dots$
4. Print alternating even and odd terms: Input the number of terms n .
 $2 1 4 3 6 5 8 7 \dots$
5. Print reciprocals of natural numbers: Input the number of terms n and display the terms as decimal values rounded to two decimal places.
 $1.00 0.50 0.33 0.25 0.20 \dots$
6. Exit the program

Accept the required input(s) based on the selected operation. Continue displaying the menu until the user selects the **Exit** option. Display an appropriate error message if the user enters an invalid menu choice. Assume that the number of terms, n , is a positive integer.

Note: Prepare the algorithm and test-case table. Then write, compile and execute the C program, and test it against the prepared test cases.